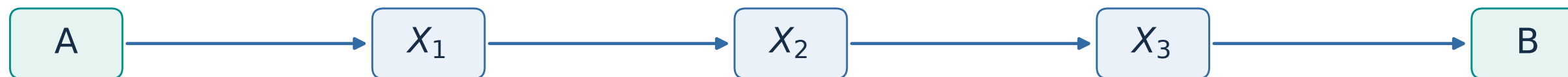


When is a reaction chain replaceable?

Repeated chain growth can sometimes be lumped into an effective step or an aggregate state.

DETAILED CHAIN / CONCEPTUAL SCHEMATIC

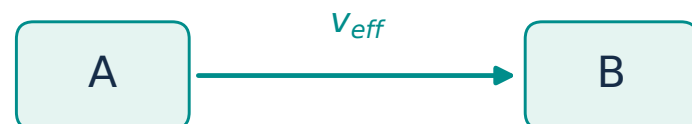


Remove internal states only after checking branches, storage, observables and stoichiometry.

Topology identifies a candidate. Chemistry and dynamics decide whether the closure is acceptable.

OPTION A

Effective $A \rightarrow B$



Negligible transit delay
No distinct storage needed
Net resource ledger retained

CONDITIONAL CANDIDATE

OPTION B

Aggregate state E

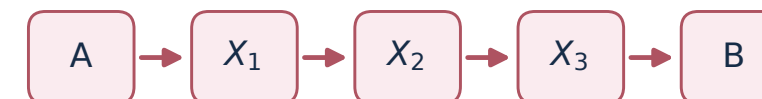


Stores total chain occupancy
Useful when transit delay matters
Requires an explicit exit closure

OCCUPANCY CANDIDATE

OPTION C

Keep explicit



Side branches or return paths
Resource timing must be tracked
Intermediate observables matter

EXPLICIT RETENTION

Local QSSA may follow either retained or aggregated motifs; topology alone does not establish timescale separation.